

INDEPENDENT PEER-REVIEW PACKET

EC-017

Positive Fixed-r Typed Finite-Q4 Palm Mass

Cold Review Packet 12 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	12 of 20
Object ID	EC-017
Object class	Finite-dimensional Gaussian/Palm positivity
Exact statement	At $r=1/20$, $P_{\{1/20, Q4\}}^{\{MS\}}(P01-ROUTE-B-D4-BOX-001) > 0$ under the finite-Q4 typed Palm law.
Internal terminal label	CLOSED - EXACT POSITIVE FIXED-r TYPED FINITE-Q4 PALM MASS OF THE ROUTE-B BOX
Terminal date	2026-07-22
Recommended expertise	Gaussian measures / Palm reweighting / interval analysis

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Interval-certified box, Gaussian nondegeneracy, finite-Q4 measure and normalizer proof, byte-exact lower-bound source, and independent law/scope review.
Prior review record	Independent review verified the qualitative positivity and covariance residual; the numerical lower bound is supporting evidence only.
Binding exclusions	No exact full-field mass, adjacency, r-uniformity, numerical terminal constant, P0.1/P0.2, or lifetime theorem.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.

- Object-specific exclusions: No exact full-field mass, adjacency, r-uniformity, numerical terminal constant, P0.1/P0.2, or lifetime theorem.

4. Object-specific hostile checks

- Verify the finite-Q4 law and exact box are identical throughout numerator and normalizer.
- Prove positive Gaussian box mass from nondegeneracy and openness.
- Check type indicators and determinant weights do not vanish on the box.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

GP-CLS-PROP-020-v1.1 — REVISED CONDITIONAL TERMINAL CLOSURE PACKAGE

POSITIVE FIXED-r TYPED FINITE-Q4 PALM MASS OF THE ROUTE-B DEGREE-FOUR BOX

Artifact ID: GP-CLS-PROP-020-v1.1

Prepared by: GP — OpenAI GPT-5.6 Thinking

Created: 2026-07-22

Class: CLS-PROP — numbered material-scope revision pending one non-GP technical review and Dylan approval

Authority: none until both gates pass

Canonical impact: NONE before approval; afterward only the exact narrow finite-Q4 object below

Queue candidate: EC-017

Supersedes for review: GP-CLS-PROP-020-v1.0, whose exact-field Palm-law wording failed GP-AUD-112

[[OBJ:GP-CLS-PROP-020-v1.1]] [[QUEUE:EC-017]] [[OBJ:GP-AUD-112-v1.0]] [[STATUS:NUMBERED-SCOPE-REVISION]]
[[STATUS:OUTSIDE-REVIEW-REQUIRED]]

1. EXACT OBJECT PROPOSED FOR CLOSURE

At the single fixed separation $r=1/20$, let γ_r be the exact normalized-periodized side-24 Gaussian law of the nine midpoint jets

$J_r=(q,a,w,z,c_{40},c_{31},c_{22},c_{13},c_{04})$

under the exact six value/gradient pins.

Let $H_M^{(4)}(J_r)$ and $H_S^{(4)}(J_r)$ be the normalized quartic endpoint Hessians defined in GP-DER-043, let

$$W_{\{r,Q4\}} = |\det H_M^{(4)} \det H_S^{(4)}|$$

$$1_{\{H_M^{(4)} \text{ negative definite}\}}$$

$$1_{\{\det H_S^{(4)} < 0\}},$$

let

$Z_{\{r,Q4\}} = E_{\{\gamma_r\}}[W_{\{r,Q4\}}],$

and define the typed finite-Q4 Palm law

$$dP_{\{r,Q4\}}^{\{MS\}} = W_{\{r,Q4\}} d\gamma_r / Z_{\{r,Q4\}}.$$

For the full-dimensional box P01-ROUTE-B-D4-BOX-001 certified in GP-DER-106,

$P_{\{1/20,Q4\}}^{\{MS\}}(P01-ROUTE-B-D4-BOX-001) > 0.$

Since every jet vector in the box belongs to the repaired degree-four capture event $H_{\{1/20\}}^{(4)}$,

$$P_{\{1/20, Q4\}^{\{MS\}}(H_{\{1/20\}^{\{4\}}})$$

$$>= P_{\{1/20, Q4\}^{\{MS\}}(P01-ROUTE B-D4-BOX-001) > 0.$$

This package closes only that finite-dimensional finite-Q4 statement.

2. WHY THIS REVISION IS REQUIRED

The P0.1 exact law weights the full conditioned field by the determinants and types of the exact endpoint Hessians.

GP-DER-043 instead constructs $W_{\{r, Q4\}}$ from normalized quartic endpoint Hessians that are functions of the nine midpoint jets. GP-DER-045 explicitly separates the exact field from its degree-four approximation and allows nonzero endpoint Jacobian remainder.

Therefore the v1.0 notation P_r^{MS} was overbroad. GP-AUD-112 records the exact object mismatch and the preserved finite-Q4 result.

3. EVIDENCE

Degree-four continuum box:

- - GP-DER-106-v1.0, Drive ID 1ve_dMzQd8QvkPpu55oReGwSRnQRWuNsbLcKWzYStGtl;
- - GP-DATA-106-v1.0 capsule, Drive ID 1sAbtS40ioX91gxiYRpbTZVg2hmQDqQu-z5HklYMibe4;
- - GP-AUD-107-v1.0, Drive ID 1cCuANEtlp1bZDvckMCNNwGBfXZivCMUWrwxwtccrRc;
- - full nine-coordinate positive-width box; all degree-four gates pass; 5/5 controls pass.

Finite-Q4 typed-Palm law and qualitative positivity:

- - GP-DER-043-v1.0, Drive ID 15CbojXsDhXh_nfYgBEI83uaddCny_OSthHqhTOsZa9M;
- - GP-DER-108-v1.0, Drive ID 1KCSuyZuIZxLajefeg-u2cgEV77vIUqdxgE8_EAp61j8;
- - GP-AUD-109-v1.0, Drive ID 1IXhB77SOFrLFZcNDRc29akY4b3k_TUICXiC1e6DW2Ms;
- - GP-DATA-035 exact finite-r nine-jet conditional Gaussian construction.

Quantitative strengthening:

- - GP-DER-110-v1.0, Drive ID 1aQDmNdn8UZf8i44o5yNcfLOyJ-wEC4inuiVLfCoVnrs;
- - GP-DATA-110-v1.0 capsule, Drive ID 18uw_WMQnxJttvwzoQd_UwohKwwPO09rJBKyUyrDiDkM;
- - GP-AUD-111-v1.0, Drive ID 198-WWjtCVBy5dfL2YvfyM0pJe8-lLFaG7O05OL-Mjp4;
- - GP-side candidate lower bound

4.728529363036503915804098882860359338485e-31;

- - 5/5 numerical-object controls pass.

Scope correction:

- - GP-AUD-112-v1.0, Drive ID 1IXo4GyjrI16Z0sI2f9i_KQDNvKZp0PhCeSILUmbY9RE.

4. EXACT MATHEMATICAL BASIS

The declared box has positive Euclidean volume in the nine-jet coordinates.

The exact normalized-periodized finite-r conditional nine-jet covariance is positive definite, so γ_r has a continuous strictly positive density on the box.

The quartic endpoint type inequalities hold strictly throughout the box, and $W_{\{r, Q4\}}$ is bounded below by a positive constant there.

The finite-Q4 normalizer $Z_{\{r,Q4\}}$ is finite and positive.

Therefore the box has strictly positive $P_{\{r,Q4\}}^{\{MS\}}$ mass.

The numerical source additionally supplies one conservative lower bound but is not required for the qualitative terminal claim.

5. INDEPENDENT REVIEW REQUIREMENT

One explicit review from CL, AO48, or CW/C047R must verify or amend:

1. that GP-DER-043 defines the finite-Q4 law used here;
2. identical nine-jet coordinate ordering and $q \leftrightarrow s$ fixed-r Jacobian;
3. exact-torus nine-jet covariance nondegeneracy at $r=1/20$;
4. full-box quartic determinant/type margins;
5. positive-volume, positive-density, and finite finite-Q4 normalizer argument;
6. the fixed-r, finite-dimensional, degree-four-only scope;
7. that no exact full-field Palm or adjacency statement is bundled.

Optional numerical review:

- - periodization-tail enclosure;
- - raw/divided-difference interval overlap;
- - density-infimum and Mahalanobis bounds;
- - Gaussian moment normalizer upper bound;
- - final 4.728529e-31 arithmetic.

Accepted technical response wording:

APPROVE EC-017 REVISION 1.1 EXACT SCOPE

The reviewer may add:

NUMERICAL LOWER BOUND ALSO VERIFIED

or

QUALITATIVE POSITIVITY APPROVED; NUMERICAL MAGNITUDE RESERVED.

Silence, GP self-review, approval of v1.0, or general agreement does not count.

6. AFFECTED DEPENDENCIES

Upon terminal approval, retire from active attention:

- - whether the full-dimensional Route-B box has positive mass under the finite-Q4 typed Palm law at $r=1/20$;
- - the fixed-r finite-Q4 positive-mass portion of GP-REQ-105 Route B;
- - repeated finite-Q4 normalizer/box-mass derivations for this exact object.

Continue actively:

- - positive mass under the exact full-field Palm law;
- - residual endpoint-Hessian/type transfer;
- - exact-field capture and adjacency A_r ;
- - conditional Gaussian support or populated GP-SCHEMA-095 repair;
- - r-interval continuation and uniform positivity;
- - P0.1, P0.2, Theorem B, elder-rule pairing, and lifetime laws;
- - optimization or outside verification of the numerical lower bound.

7. UNAFFECTED DEPENDENCIES AND EXCLUSIONS

This package does not close or imply:

- - exact full-field P_r^{MS} mass;
- - exact endpoint Hessian typing;
- - exact-field adjacency A_r ;
- - any exact-field numerical lower bound c_A ;
- - $\inf_{\{0 < r \leq r_0\}} a_r > 0$;
- - convergence as $r \rightarrow 0$;
- - the large CL-DATA-051 box or imported a_r claims;
- - P0.1 or P0.2;
- - theorem promotion, q-machine replacement, GP-AUTO-034 activation, ballot passage, release, deletion, or destructive operation.

8. PROPOSED TERMINAL LABEL

CLOSED — EXACT POSITIVE FIXED-r TYPED FINITE-Q4 PALM MASS OF THE FULL-DIMENSIONAL DEGREE-FOUR ROUTE-B CAPTURE BOX

9. REQUIRED REGISTER ACTIONS AFTER BOTH GATES

1. Append EC-017 Revision 1.1 to the Closure Log with the exact finite-Q4 object, evidence, non-GP review, exclusions, terminal label, and reopening rule.
2. Mark EC-017 TERMINAL — REMOVED FROM ACTIVE ATTENTION in the Easy Closure Queue.
3. Mark only the fixed-r finite-Q4 mass subtask of GP-REQ-105 Route B satisfied.
4. Preserve exact-field Palm mass, exact-field capture, r-uniformity, and P0.1 as active work.
5. Preserve v1.0, GP-AUD-112, all sources, receipts, and correction provenance.

10. HUMAN APPROVAL REQUIREMENT

After the non-GP technical review, Dylan M. Roy must explicitly approve:

APPROVE EC-017 REVISION 1.1 EXACT SCOPE

General authorization to continue Drive work does not substitute for this terminal decision.

11. REOPENING RULE

Reopen only by additive evidence showing a finite-Q4 coordinate/law mismatch, nine-jet covariance singularity, interval-box failure, quartic determinant/type failure, finite-Q4 Palm-weight or normalizer error, numerical source defect if the magnitude was adopted, or scope contamination.

Failure of exact-field transfer, exact full-field Palm positivity, uniform positivity, P0.1, or P0.2 does not reopen this finite-Q4 closure unless it contradicts the closed object itself.

12. CURRENT STATUS

v1.0: MATERIAL-SCOPE-FAILED / SUPERSEDED FOR REVIEW.

v1.1: COMPLETE REVISED PACKAGE.

NON-GP REVIEW: OPEN.

HUMAN APPROVAL: OPEN.

TERMINAL: NO.

END GP-CLS-PROP-020-v1.1

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT POSITIVE FIXED-r TYPED FINITE-Q4 PALM MASS OF THE ROUTE-B BOX
Closure date	2026-07-22
Evidence basis	Interval-certified box, Gaussian nondegeneracy, finite-Q4 measure and normalizer proof, byte-exact lower-bound source, and independent law/scope review.
Independent review already recorded	Independent review verified the qualitative positivity and covariance residual; the numerical lower bound is supporting evidence only.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- No exact full-field mass, adjacency, r-uniformity, numerical terminal constant, P0.1/P0.2, or lifetime theorem.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: https://docs.google.com/document/d/16LrYGWGhU-DsW5vIV5u8BaZ19gCv_KDagpg1qBAkYyI/edit

Closure / terminal record: https://docs.google.com/document/d/1-WDQtU4LpHneNztZss6ctg_lKhsyGnXdYC4Y7wM6Fc/edit

Local source snapshot used to construct this packet: GP-CLS-PROP-020-v1.1 — Revised Conditional Closure of Positive Fixed-r Finite-Q4 Palm Mass

Local source snapshot SHA-256 (DOCX export, transport checksum only):
d9c98599dd64c0a838162f4de00e67ebb1a6e16a2994f95cefb700985d45e6f0

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.